```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("world_happiness_2015_2019.xlsx")

df.head()
```

	Year	Country	Happiness Score	Economy (GDP per Capita)	Family	Health (Life Expectancy)	Freedom	Trust (Government Corruption)	Generosity
0	2015	Switzerland	7.587	1.39651	1.34951	0.94143	0.66557	0.41978	0.29678
1	2015	Iceland	7.561	1.30232	1.40223	0.94784	0.62877	0.14145	0.43630
2	2015	Denmark	7.527	1.32548	1.36058	0.87464	0.64938	0.48357	0.34139
3	2015	Norway	7.522	1.45900	1.33095	0.88521	0.66973	0.36503	0.34699
4	2015	Canada	7.427	1.32629	1.32261	0.90563	0.63297	0.32957	0.45811

后续步骤： 使用 df 生成代码 New interactive sheet

```
df.shape
```

```
(782, 9)
```

```
df.columns
```

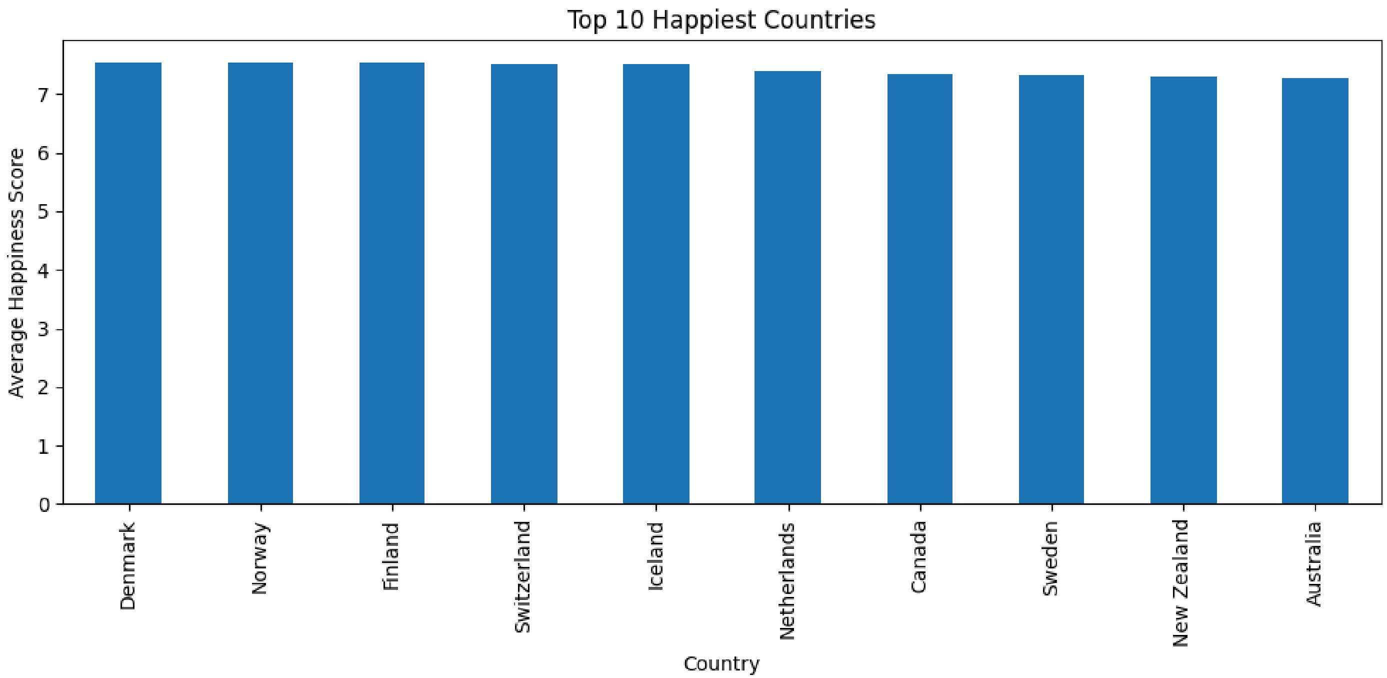
```
Index(['Year', 'Country', 'Happiness Score', 'Economy (GDP per Capita)',
       'Family', 'Health (Life Expectancy)', 'Freedom',
       'Trust (Government Corruption)', 'Generosity'],
      dtype='object')
```

```
top10 = (
        df.groupby("Country")["Happiness  Score"]
            .mean()
            .sort_values(ascending=False)
            .head(10)
)

top10
```

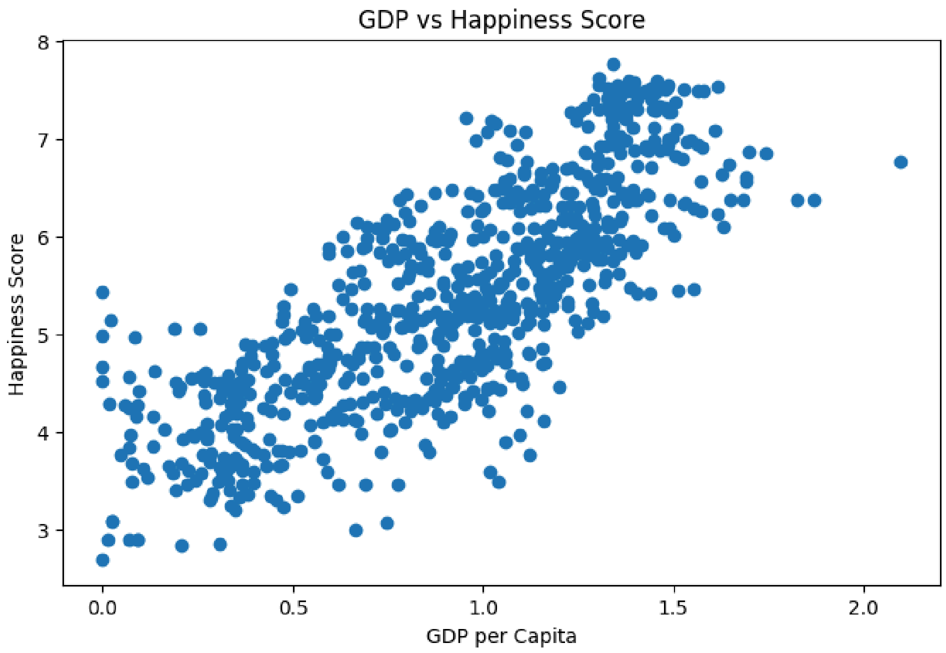
Country	Happiness Score
Denmark	7.5460
Norway	7.5410
Finland	7.5378
Switzerland	7.5114
Iceland	7.5110
Netherlands	7.4046
Canada	7.3506
Sweden	7.3192
New Zealand	7.3130
Australia	7.2762

dtype: float64

```
plt.figure(figsize=(10,5))

top10.plot(kind="bar")

plt.title("Top 10 Happiest Countries")
plt.ylabel("Average Happiness Score")

plt.tight_layout()
plt.show()
```

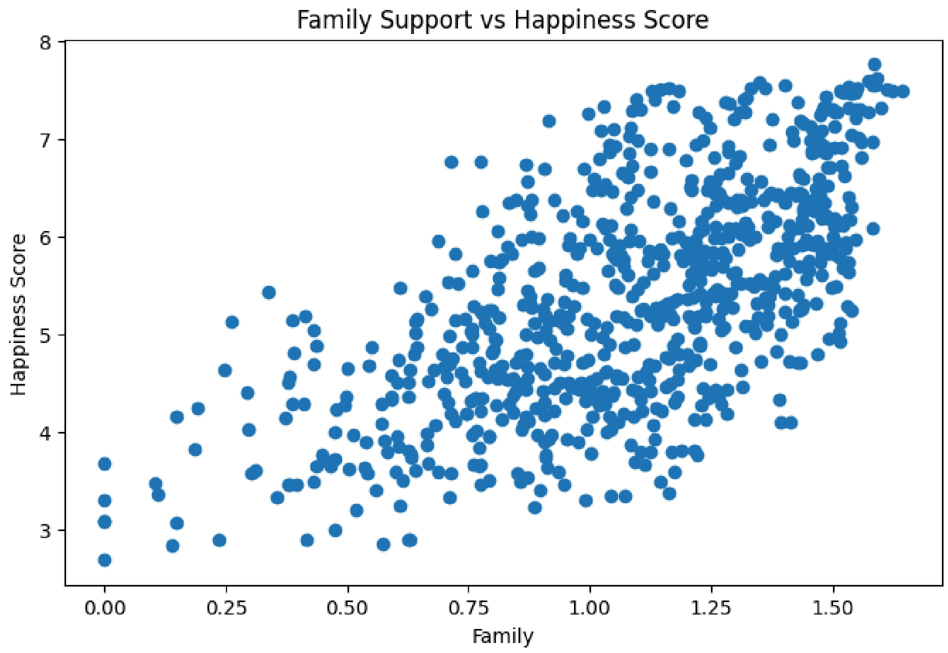



```
plt.figure(figsize=(8,5))

plt.scatter(
        df["Economy (GDP per Capita)"],
        df["Happiness Score"]
)

plt.xlabel("GDP per Capita")
plt.ylabel("Happiness Score")

plt.title("GDP vs Happiness Score")

plt.show()
```



There is a clear positive relationship between GDP per capita and happiness score. Countries with higher GDP per capita tend to have higher happiness scores.

开始借助 AI 编写或生成代码。



Family support shows a strong positive relationship with happiness score. Countries with stronger social and family support systems generally report higher levels of happiness.

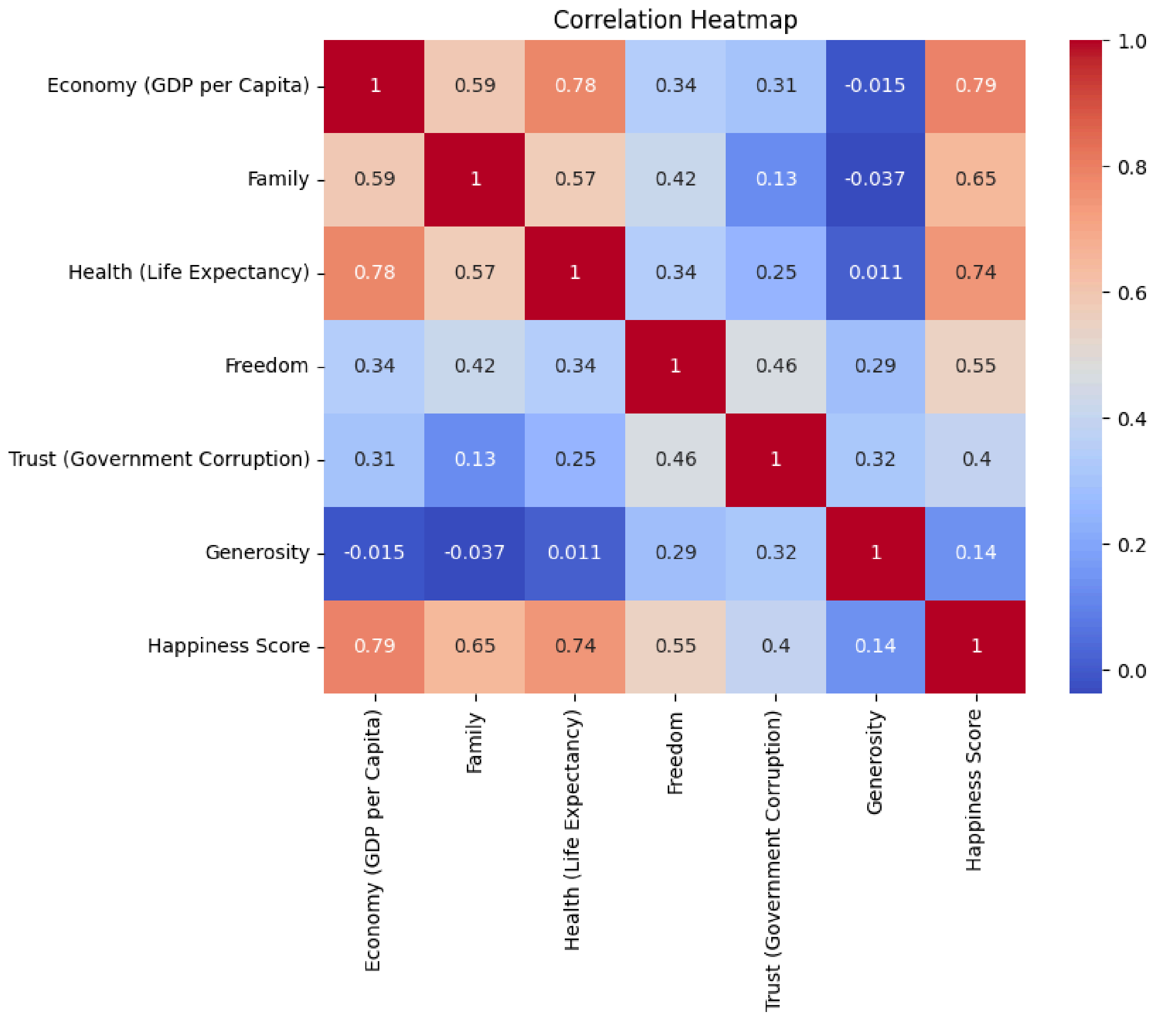
```
import seaborn as sns
```

```
features = [
        "Economy (GDP per Capita)",
        "Family",
        "Health (Life Expectancy)",
        "Freedom",
        "Trust (Government Corruption)",
        "Generosity",
        "Happiness Score"
]

corr = df[features].corr()

corr
```

	Economy (GDP per Capita)	Family	Health (Life Expectancy)	Freedom	Trust (Government Corruption)	Generosity	Happiness Score
Economy (GDP per Capita)	1.000000	0.585966	0.784338	0.340511	0.306307	-0.014560	0.789284
Family	0.585966	1.000000	0.572650	0.420361	0.126401	-0.037262	0.648799
Health (Life Expectancy)	0.784338	0.572650	1.000000	0.340745	0.250512	0.010638	0.742456
Freedom	0.340511	0.420361	0.340745	1.000000	0.459593	0.290706	0.551258
Trust (Government Corruption)	0.306307	0.126401	0.250512	0.459593	1.000000	0.318920	0.398418
Generosity	-0.014560	-0.037262	0.010638	0.290706	0.318920	1.000000	0.137578
Happiness Score	0.789284	0.648799	0.742456	0.551258	0.398418	0.137578	1.000000

后续步骤： 使用 corr 生成代码 New interactive sheet

```
plt.figure(figsize=(8,6))

sns.heatmap(
        corr,
        annot=True,
        cmap="coolwarm"
)

plt.title("Correlation  Heatmap")

plt.show()
```



conclusion1

GDP per capita has the strongest correlation with happiness score (0.79).

conclusion2

Health (Life Expectancy) is also strongly associated with happiness score (0.74).

conclusion3

Family support shows a strong positive relationship with happiness score (0.65).

conclusion4

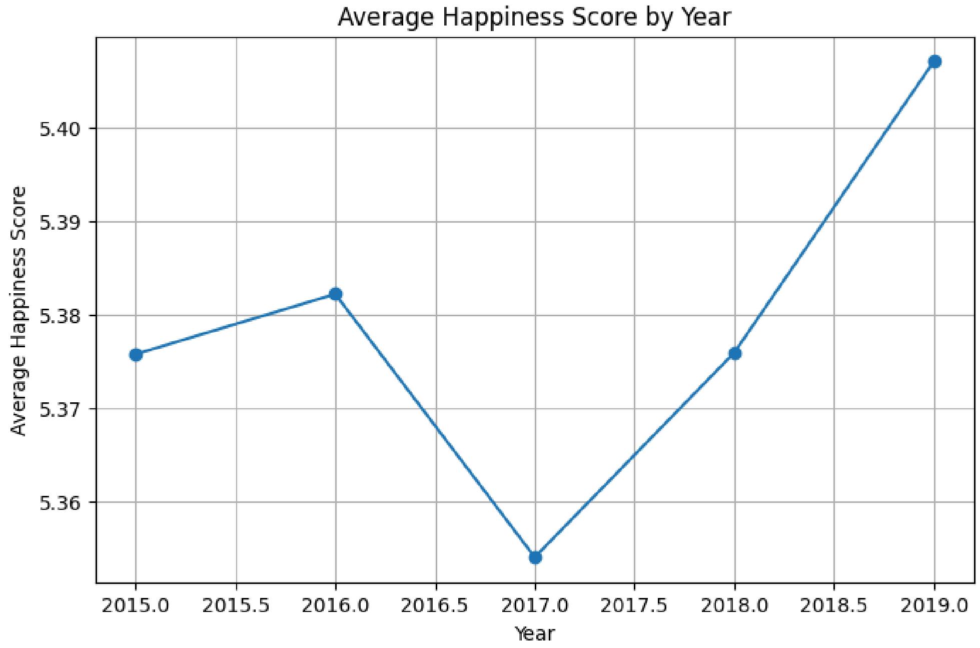
Generosity has the weakest correlation with happiness score (0.14).

```
yearly = df.groupby("Year")["Happiness Score"].mean()

yearly
```

Year	Happiness Score
2015	5.375734
2016	5.382185
2017	5.354019
2018	5.375917
2019	5.407096

dtype: float64

```
plt.figure(figsize=(8,5))

plt.plot(
        yearly.index,
        yearly.values,
        marker="o"
)

plt.title("Average Happiness Score by Year")

plt.xlabel("Year")
plt.ylabel("Average Happiness Score")

plt.grid(True)

plt.show()
```



Global average happiness scores remained relatively stable between 2015 and 2019, with a slight increase by 2019.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
```

```
df.isnull().sum()
```

0

```
df_ml = df.dropna()
```

Country 0

```
X = df_ml[
[
        "Economy (GDP per Capita)",
        "Family",
        "Health (Life Expectancy)",
        "Freedom",
        "Trust (Government Corruption)",
        "Generosity"
]
]
y = df_ml["Happiness Score"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
        X
```